THE INVOLUTION RANK OF MAPPING CLASS GROUPS OF CLOSED SURFACES

QILONG GUO AND CHUNXING YAN

ABSTRACT. For a group G admitting a finite generating set consisting of involutions, let $\operatorname{irank}(G)$ denote the least number of involutions required to generate G. For the closed orientable surface Σ_g, we prove

$$\operatorname{irank}\operatorname{MCG}(\Sigma_3) = 4, \qquad \operatorname{irank}\operatorname{MCG}(\Sigma_4) = \operatorname{irank}\operatorname{MCG}(\Sigma_5) = 3.$$

Together with the results of Korkmaz and Yıldız, this completes the determination of the involution rank for all $g \geq 3$ and gives a complete answer to Farb's problem in this range.

1. INTRODUCTION

For a group G admitting a finite generating set consisting of involutions, define its *involution rank* by

$$\operatorname{irank}(G) = \min\{n : G = \langle s_1, \ldots, s_n \rangle,\ s_i^2 = 1,\ s_i \neq 1\}.$$

Thus $\operatorname{irank}(G)$ is the minimum cardinality of a generating set consisting of involutions. Let Σ_g be a closed connected oriented surface of genus g, and let $\operatorname{MCG}(\Sigma_g)$ denote its orientation-preserving mapping class group. Farb posed the following problem [7]:

Farb's Problem 5.8. Find sharp bounds on the number of involutions needed to generate $\operatorname{MCG}(\Sigma_g)$. Equivalently, determine $\operatorname{irank}\operatorname{MCG}(\Sigma_g)$.

For $g \leq 2$, $\operatorname{MCG}(\Sigma_0)$ is trivial, while $\operatorname{MCG}(\Sigma_1)$ and $\operatorname{MCG}(\Sigma_2)$ cannot be generated by involutions [8, Chapter 5]. Korkmaz proved that four involutions suffice for every $g \geq 3$ and that three suffice for $g \geq 8$ [10]. Yıldız later reduced the latter range to $g \geq 6$ [18]. A recent survey identifies $g = 3, 4, 5$ as the only genera for which generation by three involutions remained open [3, Question 4.5]. We determine the involution rank in these three genera. The genus-three case turns out to be exceptional, whereas genera four and five already exhibit the stable behavior of the higher-genus cases. Our main theorem therefore completes the determination of the involution rank of the mapping class groups of closed orientable surfaces of genus $g \geq 3$.

Theorem 1.1.

$$\operatorname{irank}\operatorname{MCG}(\Sigma_3) = 4, \qquad \operatorname{irank}\operatorname{MCG}(\Sigma_4) = 3, \qquad \operatorname{irank}\operatorname{MCG}(\Sigma_5) = 3.$$

Together with the work of Korkmaz and Yıldız, Theorem 1.1 gives a complete answer to Farb's Problem 5.8 for closed orientable surfaces of genus $g \geq 3$.

Corollary 1.2. *For every $g \geq 3$,*

$$\operatorname{irank}\operatorname{MCG}(\Sigma_g) = \begin{cases} 4, & g = 3, \\ 3, & g \geq 4. \end{cases}$$

1991 *Mathematics Subject Classification.* Primary 57K20; Secondary 20F65.

Key words and phrases. mapping class group, involution, symplectic group, Dehn twist, minimal generation.

Related questions on generation by involutions or, more generally, by torsion elements have also been studied for punctured and extended mapping class groups; see [12, 11, 5, 9, 14, 15, 13, 2, 1, 17].

The genus-three argument differs from those in genera four and five. The mod-2 symplectic representation restricts the images of involutions to a highly constrained conjugacy class, and Scott's fixed-space inequality rules out generation by three such images. In genera four and five, we instead construct a generating pair (R, H) together with an involution ρ satisfying

$$\rho R \rho^{-1} = R^{-1}, \qquad \rho H \rho^{-1} = H^{-1}.$$

Once $\langle R, H\rangle = \mathrm{MCG}(\Sigma_g)$ is established, Lemma 2.3, together with the fact that $\mathrm{MCG}(\Sigma_g)$ cannot be generated by two involutions, shows that ρ, ρR, and ρH are three involutions generating $\mathrm{MCG}(\Sigma_g)$.

Section 2 contains the preliminaries, including the group-theoretic facts about generation by involutions, the finite-group input needed in genus three, and the cyclic curve system and fundamental-group setup used later. Section 3 proves the genus-three obstruction via the mod-2 symplectic representation. Sections 4 and 5 treat genera four and five by constructing suitable generating pairs and exploiting a reversing symmetry. Appendix A contains the explicit free-group calculations verifying the mapping-class identities used in these two sections.

2. Preliminaries

2.1. Group-theoretic preliminaries.

Lemma 2.1. *If $G = \langle s, t\rangle$ with $s^2 = t^2 = 1$, then the cyclic subgroup $\langle st\rangle$ has index at most two in G.*

Proof. Put $r = st$. Then $G = \langle s, r\rangle$ and $sr = r^{-1}s$. Hence every element of G can be written as r^n or sr^n. Therefore

$$G = \langle r\rangle \cup s\langle r\rangle,$$

and $\langle r\rangle$ has index at most two. □

Remark 2.2. For $g \geq 2$, the group $\mathrm{MCG}(\Sigma_g)$ cannot be generated by two involutions. Indeed, by Lemma 2.1, every group generated by two involutions has a cyclic subgroup of index at most two, and hence cannot contain a subgroup isomorphic to $\mathbb{Z}^2$. On the other hand, $\mathrm{MCG}(\Sigma_g)$ contains a copy of $\mathbb{Z}^2$, generated by Dehn twists about two disjoint nonisotopic essential curves.

Lemma 2.3. *Let $G = \langle r, h\rangle$. Suppose that $\rho \in G$ is an involution satisfying*

$$\rho r \rho^{-1} = r^{-1}, \qquad \rho h \rho^{-1} = h^{-1}.$$

Then ρr and ρh have order at most two, and the three elements ρ, ρr, and ρh generate G.

Proof. Since $\rho^2 = 1$,

$$(\rho r)^2 = (\rho r \rho^{-1})r = r^{-1}r = 1, \qquad (\rho h)^2 = (\rho h \rho^{-1})h = h^{-1}h = 1.$$

Moreover,

$$r = \rho(\rho r), \qquad h = \rho(\rho h),$$

so $\langle r, h\rangle \subseteq \langle \rho, \rho r, \rho h\rangle$. Since $\rho, r, h \in G$,

$$\langle \rho, \rho r, \rho h\rangle \subseteq G,$$

which proves the reverse inclusion. □

For the genus-three argument, let $G = \mathrm{Sp}(6, 2)$, the group of linear automorphisms of $\mathbb{F}_2^6$ preserving a nondegenerate alternating bilinear form. It has four conjugacy classes of involutions, conventionally denoted $2A, 2B, 2C, 2D$, with centralizer orders as follows:

class	$2A$	$2B$	$2C$	$2D$
$\lvert C_G(x)\rvert$	23040	4608	1536	384.

See the ATLAS tables [6]. Thus, for an involution $x \in G$,

$$|C_G(x)| = 4608 = 2^9 \cdot 3^2 \quad \Longrightarrow \quad x \in 2B.$$

We use Scott's fixed-space inequality in the following form.

Lemma 2.4 (Scott). *Let U be a finite-dimensional complex G-module. Suppose that*

$$G = \langle g_1, \ldots, g_r \rangle, \qquad g_1 \cdots g_r = 1.$$

Then

$$\sum_{i=1}^{r} \operatorname{codim}_U U^{g_i} \geq \operatorname{codim}_U U^G + \operatorname{codim}_{U^*} (U^*)^G. \tag{1}$$

In particular, if U is nontrivial and irreducible, then

$$\sum_{i=1}^{r} \operatorname{codim}_U U^{g_i} \geq 2 \dim U. \tag{2}$$

Proof. The inequality (1) is Scott's theorem [16, Theorem 1]. If U is nontrivial and irreducible, then

$$U^G = 0.$$

The dual module U^* is again nontrivial and irreducible, so

$$(U^*)^G = 0.$$

Hence the right-hand side of (1) is

$$\dim U + \dim U^* = 2 \dim U,$$

which gives (2). □

2.2. **Mapping-class-group preliminaries.** Let a_i, b_i, c_i denote the isotopy classes of the essential simple closed curves in Korkmaz's standard cyclic system; see [10, Figure 1]. As usual, we refer to these isotopy classes simply as curves. Let

$$\widetilde{R} : \Sigma_g \longrightarrow \Sigma_g$$

be the geometric rotation of the cyclic model through one cyclic step, an angle $2\pi/g$, and let $R = [\widetilde{R}] \in \mathrm{MCG}(\Sigma_g)$ be its mapping class. Then

$$R(a_i) = a_{i+1}, \qquad R(b_i) = b_{i+1}, \qquad R(c_i) = c_{i+1},$$

and R has order g. Choose a fixed point $p \in \Sigma_g$ of $\widetilde{R}$, which then induces an automorphism $\widetilde{R}_* \in \mathrm{Aut}(\pi_1(\Sigma_g, p))$. All subscripts on the curves and the corresponding twists are read modulo g, with representatives in $\{1, \ldots, g\}$; thus, for example, $c_0 = c_g$ and $c_{g+1} = c_1$. Throughout this paper, T_c denotes the *left Dehn twist* about c. Write

$$A_i = T_{a_i}, \qquad B_i = T_{b_i}, \qquad C_i = T_{c_i}.$$

By naturality of Dehn twists,

$$RA_iR^{-1} = A_{i+1}, \qquad RB_iR^{-1} = B_{i+1}, \qquad RC_iR^{-1} = C_{i+1}.$$

Theorem 2.5 (Korkmaz). *For every $g \geq 3$,*

$$\mathrm{MCG}(\Sigma_g) = \langle R, A_1A_2^{-1}, B_1B_2^{-1}, C_1C_2^{-1} \rangle. \tag{3}$$

Remark 2.6. Korkmaz states this theorem using right Dehn twists. Under the change from right to left Dehn twists, the three products in (3) become the inverses of those displayed there, since the two twists in each product commute. Taking inverses does not change the subgroup generated. Thus (3) is the left-twist version of [10, Theorem 5].

The following consequence of Korkmaz's generating set will be used in Sections 4 and 5.

Lemma 2.7. *Let $g \geq 3$, and let $\Gamma \leq \mathrm{MCG}(\Sigma_g)$ be a subgroup containing*

$$R, \qquad A_1B_1^{-1}, \qquad B_1C_1^{-1}, \qquad B_1C_g^{-1}.$$

Then $\Gamma = \mathrm{MCG}(\Sigma_g)$.

Proof. For indices modulo g, set

$$X_i = A_iB_i^{-1}, \qquad Y_i = B_iC_i^{-1}, \qquad Z_i = B_iC_{i-1}^{-1}.$$

The hypotheses say that $R, X_1, Y_1, Z_1 \in \Gamma$. Since conjugation by R shifts every index by one,

$$X_i, Y_i, Z_i \in \Gamma \qquad (i \in \mathbb{Z}/g\mathbb{Z}).$$

Hence

$$Y_i^{-1}Z_i = C_iC_{i-1}^{-1} \in \Gamma,$$

then

$$Y_i(C_iC_{i-1}^{-1})Y_{i-1}^{-1} = B_iB_{i-1}^{-1} \in \Gamma,$$

and finally

$$X_i(B_iB_{i-1}^{-1})X_{i-1}^{-1} = A_iA_{i-1}^{-1} \in \Gamma.$$

Taking $i = 2$ and then inverses shows that

$$A_1A_2^{-1}, \qquad B_1B_2^{-1}, \qquad C_1C_2^{-1} \in \Gamma.$$

Thus Γ contains Korkmaz's generating set in (3), so $\Gamma = \mathrm{MCG}(\Sigma_g)$. □

For the basepoint p chosen above, write $\Pi_g = \pi_1(\Sigma_g, p)$. The Dehn–Nielsen–Baer theorem gives the following identification; see, for example, [8, Chapter 8].

Lemma 2.8 (Dehn–Nielsen–Baer). *For a closed oriented surface Σ_g with $g \geq 2$, the natural action on the fundamental group induces an isomorphism*

$$\mathrm{MCG}(\Sigma_g) \cong \mathrm{Out}^+(\Pi_g), \tag{4}$$

where $\mathrm{Out}^+(\Pi_g)$ denotes the subgroup of orientation-preserving outer automorphisms. In particular, two orientation-preserving mapping classes are equal if and only if they induce the same outer automorphism of Π_g.

Choose oriented loops x_1, y_1 based at p which, after forgetting the basepoint and orientation, represent a_1 and b_1, respectively, as in Figure 1. Define

$$x_i = \widetilde{R}_*^{\,i-1}(x_1), \qquad y_i = \widetilde{R}_*^{\,i-1}(y_1), \qquad 1 \leq i \leq g.$$

We also read the indices on the based generators modulo g, so $x_{g+1} = x_1$ and $y_{g+1} = y_1$. These loops give the standard presentation

$$\Pi_g = \pi_1(\Sigma_g, p) = \left\langle x_1, y_1, \ldots, x_g, y_g \;\middle|\; \prod_{i=1}^{g} [x_i, y_i] = 1 \right\rangle, \tag{5}$$

where $[x,y] = xyx^{-1}y^{-1}$. Set

$$m_1 = x_2^{-1}y_1x_1y_1^{-1}, \qquad m_i = \widetilde{R}_*^{\,i-1}(m_1) = x_{i+1}^{-1}y_ix_iy_i^{-1}.$$

After forgetting the basepoint and orientation, the loops x_i, y_i, m_i represent a_i, b_i, c_i, respectively. Figure 1 shows the corresponding local curve configuration on the first two handles together with the based loops x_1, y_1, x_2, y_2 and m_1.

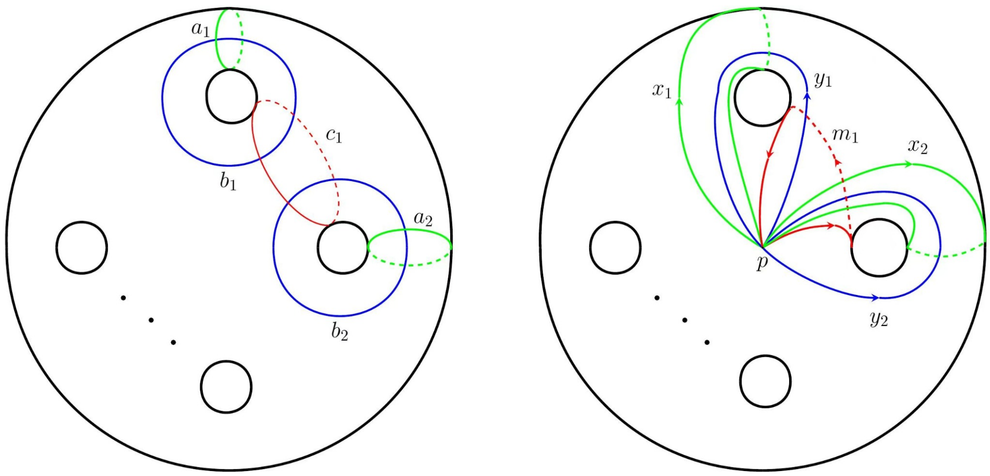

FIGURE 1. The local curve configuration on the first two handles (left) and the based loops x_1, y_1, x_2, y_2 and m_1 (right). The based loops have the common basepoint p.

With respect to these generators, the mapping classes A_i, B_i, C_i determine outer automorphisms of Π_g. We use the representatives in $\mathrm{Aut}(\Pi_g)$ listed in Table 1; the formulas follow from the standard based-loop calculation for a left Dehn twist. All indices are read modulo g.

	x_i	y_i	x_{i+1}	y_{i+1}	all other generators
A_i	x_i	$y_ix_i^{-1}$	x_{i+1}	y_{i+1}	fixed
B_i	x_iy_i	y_i	x_{i+1}	y_{i+1}	fixed
C_i	x_i	$m_i^{-1}y_i$	$m_i^{-1}x_{i+1}m_i$	$y_{i+1}m_i$	fixed

TABLE 1. Chosen representatives in $\mathrm{Aut}(\Pi_g)$ for the outer automorphisms induced by A_i, B_i, C_i. All indices are read modulo g.

For the word calculations below, we use free-group lifts of these chosen automorphisms and of $\widetilde{R}_*$. Appendix A verifies that the lifts preserve the normal closure of the surface relator and hence descend to the automorphisms above. By the Dehn–Nielsen–Baer theorem, their induced outer automorphisms correspond to the mapping classes A_i, B_i, C_i and R. We use the same notation A_i, B_i, C_i, R for the free-group lifts.

3. GENUS THREE: A SYMPLECTIC OBSTRUCTION

Consider the homomorphisms

$$\mathrm{MCG}(\Sigma_3) \xrightarrow{\Phi} \mathrm{Sp}(6,\mathbb{Z}) \xrightarrow{\pi_2} \mathrm{Sp}(6,2), \qquad \Phi_2 := \pi_2 \circ \Phi. \tag{6}$$

Here Φ is the standard symplectic representation induced by the action of the mapping class group on $H_1(\Sigma_3;\mathbb{Z})$ after choosing a symplectic basis, and it is surjective; see [8, Chapter 6]. Since π_2 is surjective, so is Φ_2. We first determine the possible images of involutions under Φ_2.

Lemma 3.1. *Let $f \in \mathrm{MCG}(\Sigma_3)$ be an involution. Then either $\Phi_2(f) = I_6$, or $\Phi_2(f)$ is conjugate in* $\mathrm{Sp}(6,2)$ *to*

$$X = \begin{pmatrix} P & 0 \\ 0 & P \end{pmatrix}, \qquad \text{where} \qquad P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}. \tag{7}$$

Proof. Set $S = \Phi(f) \in \mathrm{Sp}(6,\mathbb{Z})$. Since $f^2 = 1$, we have $S^2 = I_6$. By Braden's canonical form [4, Theorem 1.1], S is conjugate in $\mathrm{Sp}(6,\mathbb{Z})$ to

$$S_0 = \begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix}, \qquad a = I_p \oplus (-I_m) \oplus Q^{\oplus t}, \qquad p + m + 2t = 3,$$

where

$$Q = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Since reduction modulo 2 is a homomorphism, $\pi_2(S)$ is conjugate to $\pi_2(S_0)$ in $\mathrm{Sp}(6,2)$. Since $p + m + 2t = 3$, we have $t = 0$ or 1. If $t = 0$, then

$$a = I_p \oplus (-I_m), \qquad p + m = 3.$$

The reduction of a modulo 2 is I_3, and hence

$$\pi_2(S_0) = I_6.$$

Therefore $\Phi_2(f) = I_6$. If $t = 1$, then

$$a = \varepsilon \oplus Q, \qquad \varepsilon \in \{1, -1\}.$$

Since $1 = -1$ in $\mathbb{F}_2$, the reduction of a modulo 2 is

$$P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

Therefore

$$\pi_2(S_0) = \begin{pmatrix} P & 0 \\ 0 & P \end{pmatrix} = X.$$

Hence $\Phi_2(f) = \pi_2(S)$ is conjugate to X. $\square$

We next identify the involution class of X. Let

$$V = \mathbb{F}_2^6$$

be equipped with the standard symplectic form, and choose a symplectic basis

$$e_1, e_2, e_3, f_1, f_2, f_3$$

satisfying

$$\langle e_i, e_j \rangle = \langle f_i, f_j \rangle = 0, \qquad \langle e_i, f_j \rangle = \delta_{ij}.$$

With respect to this basis, X has the form (7); thus it fixes e_1, f_1 and interchanges e_2, e_3 and f_2, f_3. We determine the conjugacy class of X from the order of its centralizer.

Lemma 3.2. *Put*
$$N = X - I_V \in \operatorname{End}_{\mathbb{F}_2}(V), \qquad W = \operatorname{im} N.$$
Then
$$W = \operatorname{span}_{\mathbb{F}_2}\{e_2 + e_3, f_2 + f_3\}$$
is a two-dimensional totally isotropic subspace of V, and
$$C_{\mathrm{Sp}(6,2)}(X) = \mathrm{Stab}_{\mathrm{Sp}(6,2)}(W),$$
where
$$C_{\mathrm{Sp}(6,2)}(X) = \{g \in \mathrm{Sp}(6,2) : gX = Xg\}, \qquad \mathrm{Stab}_{\mathrm{Sp}(6,2)}(W) = \{g \in \mathrm{Sp}(6,2) : g(W) = W\}.$$

Proof. Set
$$w_1 = e_2 + e_3, \qquad w_2 = f_2 + f_3.$$
From the action of X we have
$$N(e_1) = N(f_1) = 0, \qquad N(e_2) = N(e_3) = w_1, \qquad N(f_2) = N(f_3) = w_2.$$
Hence
$$W = \operatorname{im} N = \operatorname{span}_{\mathbb{F}_2}\{w_1, w_2\}.$$
The vectors w_1, w_2 are linearly independent, so $\dim W = 2$. Moreover,
$$\langle w_1, w_1\rangle = 0, \qquad \langle w_2, w_2\rangle = 0, \qquad \langle w_1, w_2\rangle = \langle e_2 + e_3, f_2 + f_3\rangle = 1 + 1 = 0.$$
Hence W is totally isotropic.

It remains to prove
$$C_{\mathrm{Sp}(6,2)}(X) = \mathrm{Stab}_{\mathrm{Sp}(6,2)}(W).$$

First let $g \in C_{\mathrm{Sp}(6,2)}(X)$. Since $N = X - I_V$, we also have $gN = Ng$. As g is invertible,
$$g(W) = g(N(V)) = N(g(V)) = N(V) = W,$$
so $g \in \mathrm{Stab}_{\mathrm{Sp}(6,2)}(W)$.

Conversely, let $g \in \mathrm{Stab}_{\mathrm{Sp}(6,2)}(W)$. Since $X = I_V + N$, it is enough to prove $gNg^{-1} = N$. For
$$v = \sum_{i=1}^{3} \alpha_i e_i + \sum_{i=1}^{3} \beta_i f_i,$$
we have
$$N(v) = (\alpha_2 + \alpha_3)w_1 + (\beta_2 + \beta_3)w_2.$$
Since
$$\langle v, w_2\rangle = \alpha_2 + \alpha_3, \qquad \langle v, w_1\rangle = \beta_2 + \beta_3,$$
it follows that
$$N(v) = \langle v, w_2\rangle w_1 + \langle v, w_1\rangle w_2. \tag{8}$$
Since $g(W) = W$, there are $a, b, c, d \in \mathbb{F}_2$ such that
$$g(w_1) = aw_1 + cw_2, \qquad g(w_2) = bw_1 + dw_2.$$
The restriction $g|_W$ is invertible, so $ad + bc = 1$. Since g preserves the symplectic form, for every $v \in V$ we have
$$\begin{aligned} gNg^{-1}(v) &= \langle v, g(w_2)\rangle g(w_1) + \langle v, g(w_1)\rangle g(w_2) \\ &= (ad + bc)\big(\langle v, w_2\rangle w_1 + \langle v, w_1\rangle w_2\big) \\ &= N(v). \end{aligned}$$

Hence $gNg^{-1} = N$, and therefore $gXg^{-1} = X$. Thus $g \in C_{\mathrm{Sp}(6,2)}(X)$, proving the reverse inclusion. $\square$

Lemma 3.3. *If $f \in \mathrm{MCG}(\Sigma_3)$ is an involution, then $\Phi_2(f)$ is either trivial or belongs to the ATLAS class $2B$ of $\mathrm{Sp}(6,2)$.*

Proof. By Lemma 3.1, it remains only to prove $X \in 2B$. Let $\mathcal{I}$ denote the set of two-dimensional totally isotropic subspaces of V. By Witt's lemma, $\mathrm{Sp}(6,2)$ acts transitively on $\mathcal{I}$. Hence, for $W \in \mathcal{I}$, the orbit–stabilizer theorem gives

$$|\operatorname{Sp}(6,2)| = |\mathcal{I}|\,|\operatorname{Stab}_{\mathrm{Sp}(6,2)}(W)|.$$

Let $\mathcal{B}$ be the set of ordered bases of the subspaces in $\mathcal{I}$. Each member of $\mathcal{I}$ has

$$|\operatorname{GL}(2,2)| = (2^2-1)(2^2-2) = 6$$

ordered bases, so

$$|\mathcal{B}| = 6|\mathcal{I}|.$$

To compute $|\mathcal{B}|$, choose the first basis vector u arbitrarily among the $2^6 - 1 = 63$ nonzero vectors of V. For fixed u, the second basis vector must lie in

$$u^{\perp} \setminus \operatorname{span}_{\mathbb{F}_2}\{u\}.$$

Since $|u^{\perp}| = 2^5 = 32$, there are $32 - 2 = 30$ choices. Hence

$$|\mathcal{B}| = 63 \cdot 30,$$

and therefore

$$|\mathcal{I}| = \frac{|\mathcal{B}|}{6} = \frac{63 \cdot 30}{6} = 315.$$

Since $|\operatorname{Sp}(6,2)| = 1451520$, we obtain

$$|\operatorname{Stab}_{\mathrm{Sp}(6,2)}(W)| = \frac{|\operatorname{Sp}(6,2)|}{|\mathcal{I}|} = \frac{1451520}{315} = 4608.$$

By Lemma 3.2,

$$|C_{\mathrm{Sp}(6,2)}(X)| = |\operatorname{Stab}_{\mathrm{Sp}(6,2)}(W)| = 4608.$$

Since $|C_{\mathrm{Sp}(6,2)}(X)| = 4608$, the ATLAS data in Section 2.1 give $X \in 2B$. $\square$

Lemma 3.4. *The group $G = \mathrm{Sp}(6,2)$ cannot be generated by three elements of $\{1\} \cup 2B$.*

Proof. The ATLAS character table [6] contains an irreducible character χ of degree 15 with

$$\chi(2B) = 7.$$

Let U be a corresponding irreducible complex representation. If $x \in 2B$, then $x^2 = 1$, so every eigenvalue of x on U is ± 1. Put $a = \dim U^x$. The (-1)-eigenspace then has dimension $15 - a$, and therefore

$$\chi(x) = a - (15 - a) = 2a - 15.$$

Hence

$$\dim U^x = \frac{15 + \chi(x)}{2} = \frac{15 + 7}{2} = 11, \qquad \operatorname{codim}_U U^x = 4.$$

Suppose, to the contrary, that

$$G = \langle x_1, x_2, x_3 \rangle, \qquad x_i \in \{1\} \cup 2B \quad (1 \le i \le 3).$$

If $x_i = 1$, then $U^{x_i} = U$. Therefore

$$\operatorname{codim}_U U^{x_i} \le 4 \qquad (1 \le i \le 3). \tag{9}$$

Set

$$x_4 = (x_1x_2x_3)^{-1}.$$

Then

$$x_1x_2x_3x_4 = 1, \qquad \langle x_1, x_2, x_3, x_4\rangle = G.$$

By Lemma 2.4,

$$30 = 2\dim U \le \sum_{i=1}^{4} \operatorname{codim}_U U^{x_i}.$$

On the other hand, (9) and the trivial bound

$$\operatorname{codim}_U U^{x_4} \le \dim U = 15$$

give

$$\sum_{i=1}^{4} \operatorname{codim}_U U^{x_i} \le 4+4+4+15 = 27,$$

a contradiction. ☐

Theorem 3.5. *The mapping class group* $\mathrm{MCG}(\Sigma_3)$ *cannot be generated by three involutions. Consequently*

$$\operatorname{irank}\mathrm{MCG}(\Sigma_3) = 4.$$

Proof. Suppose

$$\mathrm{MCG}(\Sigma_3) = \langle f_1, f_2, f_3\rangle, \qquad f_i^2 = 1.$$

Since $\Phi_2 : \mathrm{MCG}(\Sigma_3) \twoheadrightarrow \mathrm{Sp}(6,2)$ is surjective,

$$\mathrm{Sp}(6,2) = \langle \Phi_2(f_1), \Phi_2(f_2), \Phi_2(f_3)\rangle.$$

By Lemma 3.3,

$$\Phi_2(f_i) \in \{1\} \cup 2B \qquad (1 \le i \le 3),$$

contradicting Lemma 3.4. Thus $\operatorname{irank}\mathrm{MCG}(\Sigma_3) \ge 4$. Korkmaz proved that four involutions generate $\mathrm{MCG}(\Sigma_g)$ for every $g \ge 3$ [10, Theorem 1], so equality holds. ☐

4. Genus four: a generating pair

The arguments in genera four and five are parallel. In each case we construct an element H_g reversed by a half-turn ρ, while Appendix A provides the three elements required by Lemma 2.7. We begin with genus four.

For genus four, let $\rho \in \mathrm{MCG}(\Sigma_4)$ be the mapping class represented by the orientation-preserving half-turn shown in Figure 2. It acts on the indicated curve classes by

$$\rho(a_i) = a_{5-i}, \qquad \rho(b_i) = b_{5-i}, \qquad \rho(c_i) = c_{4-i}, \tag{10}$$

with indices modulo four.

The symmetry in Figure 2 gives

$$\rho R \rho^{-1} = R^{-1}. \tag{11}$$

By Lemma 2.3, it remains to construct $H_4 \in \mathrm{MCG}(\Sigma_4)$ such that $\langle R, H_4\rangle = \mathrm{MCG}(\Sigma_4)$ and $\rho H_4 \rho^{-1} = H_4^{-1}$. Set

$$q_4 = C_1^{-1}B_1^{-1}A_1^{-1}A_2B_2^{-1}A_2^{-1}, \tag{12}$$

and

$$H_4 = q_4\rho q_4^{-1}\rho^{-1}. \tag{13}$$

Since $\rho^2 = 1$, one has $\rho H_4 \rho^{-1} = H_4^{-1}$.

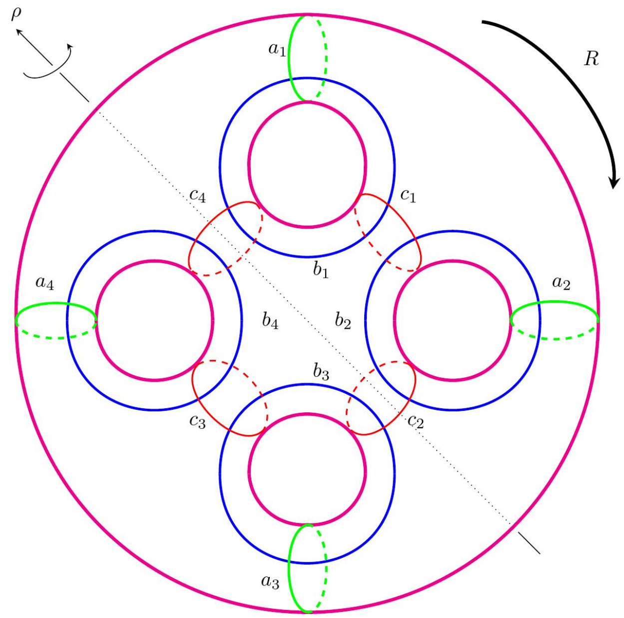

FIGURE 2. The genus-four cyclic curve system, with geometric representatives of the mapping classes R and ρ.

Lemma 4.1. *One has*

$$A_1B_1^{-1}, \qquad B_1C_1^{-1}, \qquad B_1C_4^{-1} \in \langle R, H_4\rangle.$$

Proof. Appendix A verifies

$$A_1B_1^{-1}, \qquad B_1C_1^{-1}, \qquad B_4C_3^{-1} \in \langle R, H_4\rangle.$$

Conjugating the last element by R gives $B_1C_4^{-1}$. □

Theorem 4.2. *The three elements*

$$\rho, \qquad \rho R, \qquad \rho H_4$$

are involutions and generate $\mathrm{MCG}(\Sigma_4)$. *Consequently*

$$\operatorname{irank} \mathrm{MCG}(\Sigma_4) = 3.$$

Proof. Set

$$\Gamma = \langle R, H_4\rangle.$$

By Lemma 4.1, the hypotheses of Lemma 2.7 are satisfied, and hence

$$\Gamma = \mathrm{MCG}(\Sigma_4).$$

The relations (11) and $\rho H_4 \rho^{-1} = H_4^{-1}$ allow us to apply Lemma 2.3. Thus ρR and ρH_4 have order at most two, and ρ, ρR, and ρH_4 generate $\mathrm{MCG}(\Sigma_4)$. Neither ρR nor ρH_4 is trivial, for otherwise $\mathrm{MCG}(\Sigma_4)$ would be generated by at most two involutions, contrary to Remark 2.2. Hence all three generators are involutions. Remark 2.2 also gives $\operatorname{irank} \mathrm{MCG}(\Sigma_4) \geq 3$, completing the proof. □

5. Genus five: a generating pair

For genus five, let $\rho \in \mathrm{MCG}(\Sigma_5)$ be the mapping class represented by the orientation-preserving half-turn shown in Figure 3. It acts on the indicated curve classes by

$$\rho(a_i) = a_{2-i}, \qquad \rho(b_i) = b_{2-i}, \qquad \rho(c_i) = c_{1-i}, \tag{14}$$

with indices modulo five.

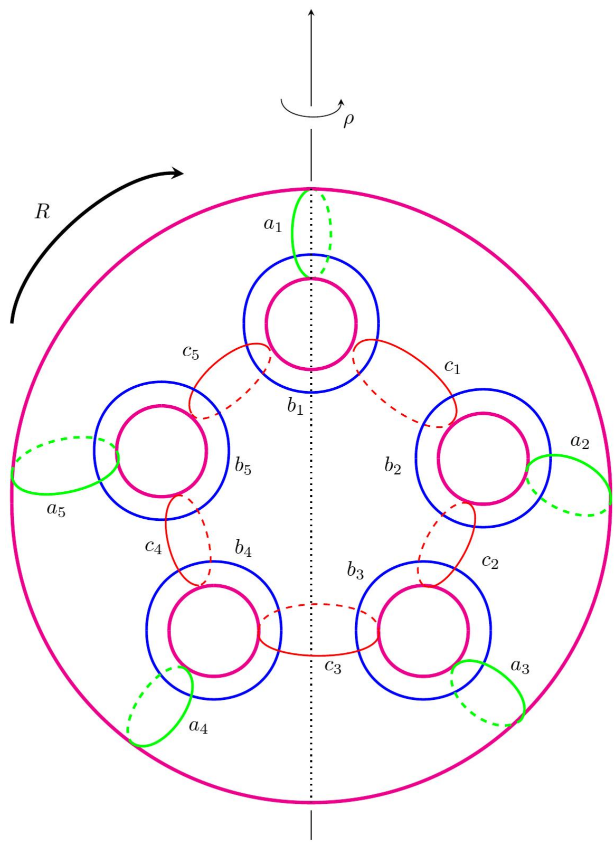

FIGURE 3. The genus-five cyclic curve system, with geometric representatives of the mapping classes R and ρ.

The symmetry in Figure 3 gives

$$\rho R \rho^{-1} = R^{-1}. \tag{15}$$

Set

$$q_5 = A_2 B_5 C_2, \tag{16}$$

and

$$H_5 = q_5 \rho q_5^{-1} \rho^{-1}. \tag{17}$$

As in genus four, $\rho^2 = 1$ gives $\rho H_5 \rho^{-1} = H_5^{-1}$.

Lemma 5.1. *One has*

$$A_1 B_1^{-1}, \qquad B_1 C_1^{-1}, \qquad B_1 C_5^{-1} \in \langle R, H_5 \rangle.$$

Proof. Appendix A gives explicit words in R and H_5 and the corresponding free-group reductions establishing these three inclusions. $\square$

Theorem 5.2. *The three elements*
$$\rho, \qquad \rho R, \qquad \rho H_5$$
are involutions and generate $\mathrm{MCG}(\Sigma_5)$*. Consequently*
$$\operatorname{irank} \mathrm{MCG}(\Sigma_5) = 3.$$

Proof. Set
$$\Gamma = \langle R, H_5 \rangle.$$
By Lemma 5.1, the hypotheses of Lemma 2.7 are satisfied, and hence
$$\Gamma = \mathrm{MCG}(\Sigma_5).$$
The relations (15) and $\rho H_5 \rho^{-1} = H_5^{-1}$ allow us to apply Lemma 2.3. Thus ρR and ρH_5 have order at most two, and ρ, ρR, and ρH_5 generate $\mathrm{MCG}(\Sigma_5)$. Neither ρR nor ρH_5 is trivial, for otherwise $\mathrm{MCG}(\Sigma_5)$ would be generated by at most two involutions, contrary to Remark 2.2. Hence all three generators are involutions. Remark 2.2 also gives $\operatorname{irank} \mathrm{MCG}(\Sigma_5) \geq 3$, completing the proof. □

Proof of Corollary 1.2. Combine Theorems 3.5, 4.2, and 5.2 with Yıldız's theorem for $g \geq 6$ [18] and Remark 2.2. □

ACKNOWLEDGEMENTS

Use of AI. The authors used OpenAI's ChatGPT to assist with the organization and exposition of the manuscript and the identification of related literature. The authors independently verified all mathematical arguments and references in the final version and take full responsibility for the content of the manuscript.

APPENDIX A. EXPLICIT WORD CALCULATIONS

We verify the identities used in Lemmas 4.1 and 5.1 by calculations in the free group
$$F_g = F(x_1, y_1, \ldots, x_g, y_g).$$
The surface group has the presentation
$$\Pi_g \cong F_g / \langle\langle r_g \rangle\rangle, \qquad \text{where } r_g = \prod_{i=1}^{g} [x_i, y_i].$$
The following lemma records the basic free-group facts used below.

Lemma. *The maps A_i, B_i, C_i given by Table 1, interpreted on the free generators, together with*
$$R(x_i) = x_{i+1}, \qquad R(y_i) = y_{i+1},$$
define automorphisms of F_g. Each of them preserves the normal closure $\langle\langle r_g \rangle\rangle$, and hence descends to an automorphism of Π_g.

Proof. For A_i and B_i, the inverses are obtained from the formulas
$$y_i \mapsto y_i x_i, \qquad x_i \mapsto x_i y_i^{-1},$$
respectively. For C_i, one checks that $C_i(m_i) = m_i$; hence its inverse is given on the affected generators by
$$y_i \mapsto m_i y_i, \qquad x_{i+1} \mapsto m_i x_{i+1} m_i^{-1}, \qquad y_{i+1} \mapsto y_{i+1} m_i^{-1}.$$
The map R cyclically permutes the free generators and is therefore an automorphism.

Moreover, A_i and B_i preserve $[x_i, y_i]$ and fix all other generators, so
$$A_i(r_g) = r_g, \qquad B_i(r_g) = r_g.$$

For C_i, cyclically rewrite r_g so that it begins with

$$[x_i, y_i][x_{i+1}, y_{i+1}].$$

This cyclic conjugate has the same normal closure as r_g. The automorphism C_i fixes every remaining commutator $[x_j, y_j]$ with $j \neq i, i+1$, while direct substitution gives

$$C_i\big([x_i, y_i][x_{i+1}, y_{i+1}]\big) = [x_i, y_i][x_{i+1}, y_{i+1}].$$

Thus C_i preserves $\langle\langle r_g \rangle\rangle$. Finally,

$$R(r_g) = [x_1, y_1]^{-1} r_g [x_1, y_1],$$

so R preserves the same normal closure. The claimed descent to Π_g follows. □

By the preceding lemma, all free-group automorphisms used below descend to automorphisms of Π_g. Consequently, an identity between words in these automorphisms can be checked at the free-group level: it suffices to verify that the two sides have the same freely reduced image on each generator

$$x_1, y_1, \ldots, x_g, y_g.$$

Equality on the free generators gives equality in $\mathrm{Aut}(F_g)$, hence the same induced automorphism of Π_g and the same element of $\mathrm{Out}^+(\Pi_g)$. By Lemma 2.8, the corresponding mapping classes are therefore equal.

Accordingly, the calculations below list the images of the nonfixed generators and indicate the remaining generators as fixed. Every comparison is made by free reduction; the surface relation is never used.

A.1. **Genus four.** To verify Lemma 4.1, recall that

$$q_4 = C_1^{-1} B_1^{-1} A_1^{-1} A_2 B_2^{-1} A_2^{-1}, \qquad H_4 = q_4 \rho q_4^{-1} \rho^{-1}.$$

Thus

$$q_4^{-1} = A_2 B_2 A_2^{-1} A_1 B_1 C_1.$$

Since ρ is orientation preserving,

$$\rho T_c \rho^{-1} = T_{\rho(c)}.$$

Using (10),

$$\begin{aligned} \rho A_1 \rho^{-1} &= A_4, & \rho A_2 \rho^{-1} &= A_3, \\ \rho B_1 \rho^{-1} &= B_4, & \rho B_2 \rho^{-1} &= B_3, \\ \rho C_1 \rho^{-1} &= C_3. \end{aligned}$$

Hence $\rho A_2^{-1} \rho^{-1} = A_3^{-1}$, and therefore

$$\rho q_4^{-1} \rho^{-1} = A_3 B_3 A_3^{-1} A_4 B_4 C_3,$$

and consequently

$$H_4 = C_1^{-1} B_1^{-1} A_1^{-1} A_2 B_2^{-1} A_2^{-1} A_3 B_3 A_3^{-1} A_4 B_4 C_3.$$

Put

$$\Gamma = \langle R, H_4 \rangle, \qquad K = H_4 (R^2 H_4 R^{-2}), \qquad M(h) = h K h^{-1} \quad (h \in \Gamma).$$

The freely reduced action of K is

$$\begin{aligned} K(x_1) &= x_1, & K(y_1) &= y_1x_1, \\ K(x_2) &= x_2, & K(y_2) &= y_2x_2^{-1}, \\ K(x_3) &= x_3y_3^{-1}, & K(y_3) &= y_3, \\ K(x_4) &= x_4y_4, & K(y_4) &= y_4. \end{aligned}$$

Comparing this action with Table 1, we see that it is exactly the action of $A_1^{-1}A_2B_3^{-1}B_4$. Hence

$$K = A_1^{-1}A_2B_3^{-1}B_4$$

in $\mathrm{Aut}(F_4)$.

We now verify the three inclusions required in Lemma 4.1.

The element $A_1B_1^{-1}$. Define

$$u_1 = H_4^2RH_4^{-1}RH_4R, \qquad u_2 = H_4^3RH_4RH_4^{-1}R^{-1}.$$

Free reduction of $M(u_1)^{-1}M(u_2)$ gives

$$x_1 \longmapsto x_1^2y_1^{-1}, \quad y_1 \longmapsto y_1x_1^{-1},$$

and it fixes $x_2, y_2, x_3, y_3, x_4, y_4$. Comparing with Table 1, this is exactly the action of $A_1B_1^{-1}$. Hence

$$M(u_1)^{-1}M(u_2) = A_1B_1^{-1} \in \Gamma.$$

The element $B_1C_1^{-1}$. Define

$$v_1 = H_4^{-2}RH_4R^{-1}H_4^{-1}R, \qquad v_2 = H_4^{-2}R^2H_4^{-1}R^{-1}H_4^{-1}R^{-1}H_4R^{-1}.$$

The freely reduced action of $M(v_1)M(v_2)^{-1}$ on the nonfixed generators is

$$\begin{aligned} x_1 &\longmapsto x_1y_1, \\ y_1 &\longmapsto x_2^{-1}y_1x_1y_1, \\ x_2 &\longmapsto x_2^{-1}y_1x_1x_2x_1^{-1}y_1^{-1}x_2, \\ y_2 &\longmapsto y_2x_1^{-1}y_1^{-1}x_2, \end{aligned}$$

and it fixes x_3, y_3, x_4, y_4. Comparing with Table 1, this is exactly the action of $B_1C_1^{-1}$. Hence

$$M(v_1)M(v_2)^{-1} = B_1C_1^{-1} \in \Gamma.$$

The element $B_1C_4^{-1}$. Define

$$w_1 = H_4^2R^{-1}H_4^{-1}RH_4R, \qquad w_2 = H_4^2R^2H_4RH_4RH_4^{-1}R^{-1}.$$

Free reduction of $M(w_1)^{-1}M(w_2)$ gives

$$\begin{aligned} y_3 &\longmapsto y_4^{-1}x_4^{-1}y_3x_3, \\ x_4 &\longmapsto y_4^{-1}x_4^{-1}y_3x_3y_3^{-1}x_4y_4y_3x_3^{-1}y_3^{-1}x_4y_4, \\ y_4 &\longmapsto y_4y_3x_3^{-1}y_3^{-1}x_4y_4, \end{aligned}$$

and it fixes x_1, y_1, x_2, y_2, x_3. Comparing with Table 1, this is exactly the action of $B_4C_3^{-1}$. Hence

$$M(w_1)^{-1}M(w_2) = B_4C_3^{-1} \in \Gamma.$$

Conjugating by R yields

$$B_1C_4^{-1} \in \Gamma.$$

This proves Lemma 4.1.

A.2. **Genus five.** For Lemma 5.1, recall that

$$q_5 = A_2B_5C_2, \qquad H_5 = q_5\rho q_5^{-1}\rho^{-1}.$$

Since ρ is orientation preserving, (14) gives

$$\rho A_2\rho^{-1} = A_5, \qquad \rho B_5\rho^{-1} = B_2, \qquad \rho C_2\rho^{-1} = C_4.$$

Hence

$$\rho q_5\rho^{-1} = A_5B_2C_4.$$

The curves a_5, b_2, c_4 are pairwise disjoint, so these twists commute. Therefore

$$\rho q_5^{-1}\rho^{-1} = A_5^{-1}B_2^{-1}C_4^{-1},$$

and consequently

$$H_5 = A_2B_5C_2A_5^{-1}B_2^{-1}C_4^{-1}.$$

Put

$$\Gamma = \langle R, H_5\rangle, \qquad H_{5,i} = R^{i-1}H_5R^{1-i}.$$

The required inclusions will follow from three auxiliary constructions.

The family $D_i = A_iA_{i+1}^{-1}$. Define

$$D_1 = H_{5,3}H_{5,2}^{-1}H_{5,3}^{-1}H_{5,5}^{-1}H_{5,1}^{-1}H_{5,5}H_{5,4}^{-1}. \tag{18}$$

Its freely reduced action is

$$\begin{aligned} D_1(x_i) &= x_i && (1 \le i \le 5),\\ D_1(y_1) &= y_1x_1^{-1}, \quad D_1(y_2) = y_2x_2, &&\\ D_1(y_i) &= y_i && (3 \le i \le 5). \end{aligned}$$

Comparing with Table 1, these are exactly the images under $A_1A_2^{-1}$. Hence

$$D_1 = A_1A_2^{-1} \in \Gamma. \tag{19}$$

For indices modulo five, set

$$D_i = R^{i-1}D_1R^{1-i}.$$

By (19) and naturality under the rotation,

$$D_i = A_iA_{i+1}^{-1} \in \Gamma \qquad (i \in \mathbb{Z}/5\mathbb{Z}). \tag{20}$$

The element $B_1A_3^{-1}$. Set

$$v = H_{5,2}D_1^{-1}H_{5,2}^{-1}.$$

The freely reduced action of v is

$$\begin{aligned} v(x_1) &= y_1^{-1}, \quad v(y_1) = y_1x_1y_1,\\ v(x_2) &= x_2, \quad\ \ v(y_2) = y_2x_2^{-1},\\ v(x_i) &= x_i, \quad\ \ v(y_i) = y_i \qquad (3 \le i \le 5). \end{aligned}$$

Thus, viewed as a mapping class via the Dehn–Nielsen–Baer theorem, v sends a_1 to b_1 and fixes a_3. Since $D_1D_2 = A_1A_3^{-1}$, naturality of Dehn twists gives

$$v(D_1D_2)v^{-1} = B_1A_3^{-1} \in \Gamma. \tag{21}$$

As a direct check, free reduction of the left-hand side gives

$$x_1 \longmapsto x_1y_1, \quad y_3 \longmapsto y_3x_3,$$

with all other free generators fixed.

The element $C_5A_4^{-1}$. Define

$$z = D_3\big(v(D_1D_2)v^{-1}\big)(D_1D_2)^{-1}\big((D_2H_{5,1}^{-1})D_1(D_2H_{5,1}^{-1})^{-1}\big)H_{5,4}D_4. \tag{22}$$

Its freely reduced action is

$$\begin{aligned}
z(x_1) &= y_5x_5^{-1}y_5^{-1}x_1y_1y_5x_5y_5^{-1},\\
z(y_1) &= x_1^{-1}y_5x_5y_5^{-1},\\
z(x_2) &= x_2y_2^{-1},\\
z(y_2) &= x_3y_3^{-1}x_3^{-1}y_2x_2y_2^{-1},\\
z(x_3) &= x_3y_3^{-1}x_3^{-1}y_2x_2y_2^{-2}x_3y_3x_3^{-1}y_2^2x_2^{-1}y_2^{-1}x_3y_3x_3^{-1},\\
z(y_3) &= y_3^2x_3^{-1}y_2^2x_2^{-1}y_2^{-1}x_3y_3x_3^{-1},\\
z(x_4) &= x_4, \qquad\qquad z(y_4) = y_4,\\
z(x_5) &= x_5^2y_5^{-1}y_1^{-1}x_1^{-1}y_5x_5y_5^{-1},\\
z(y_5) &= y_5x_5^{-1}y_5^{-1}x_1y_1y_5x_5y_5^{-1}y_1^{-1}x_1^{-1}y_5x_5y_5^{-1}.
\end{aligned}$$

In particular,

$$z(y_1) = m_5, \qquad z(x_4) = x_4.$$

Thus, viewed as a mapping class via the Dehn–Nielsen–Baer theorem, z sends b_1 to c_5 and fixes a_4. By (21) and (20),

$$v(D_1D_2)v^{-1}D_3 = B_1A_4^{-1},$$

so conjugating by z yields

$$z\big(v(D_1D_2)v^{-1}D_3\big)z^{-1} = C_5A_4^{-1} \in \Gamma. \tag{23}$$

The freely reduced action of the left-hand side on the nonfixed generators is

$$\begin{aligned}
x_1 &\longmapsto y_5x_5^{-1}y_5^{-1}x_1y_5x_5y_5^{-1},\\
y_1 &\longmapsto y_1x_1^{-1}y_5x_5y_5^{-1},\\
y_4 &\longmapsto y_4x_4,\\
y_5 &\longmapsto y_5x_5^{-1}y_5^{-1}x_1y_5,
\end{aligned}$$

and it fixes $x_2, y_2, x_3, y_3, x_4, x_5$.

Combining these auxiliary identities gives the three elements required in Lemma 5.1. From (20), (21), and (23),

$$\begin{aligned}
A_1B_1^{-1} &= (D_1D_2)(B_1A_3^{-1})^{-1} \in \Gamma,\\
B_1C_1^{-1} &= (B_1A_3^{-1})(D_3D_4)\big(R(C_5A_4^{-1})R^{-1}\big)^{-1} \in \Gamma,\\
B_1C_5^{-1} &= (B_1A_3^{-1})D_3(C_5A_4^{-1})^{-1} \in \Gamma.
\end{aligned}$$

The following table gives a direct free-group check of these three final identities:

	nontrivial images	all other generators
$A_1B_1^{-1}$	$x_1 \mapsto x_1^2 y_1^{-1}, \quad y_1 \mapsto y_1 x_1^{-1}$	fixed
$B_1C_1^{-1}$	$x_1 \mapsto x_1 y_1,$ $y_1 \mapsto x_2^{-1} y_1 x_1 y_1,$ $x_2 \mapsto x_2^{-1} y_1 x_1 x_2 x_1^{-1} y_1^{-1} x_2,$ $y_2 \mapsto y_2 x_1^{-1} y_1^{-1} x_2$	fixed
$B_1C_5^{-1}$	$x_1 \mapsto y_1^{-1} x_1^{-1} y_5 x_5 y_5^{-1} x_1 y_1 y_5 x_5^{-1} y_5^{-1} x_1 y_1,$ $y_1 \mapsto y_1 y_5 x_5^{-1} y_5^{-1} x_1 y_1,$ $y_5 \mapsto y_1^{-1} x_1^{-1} y_5 x_5$	fixed

This proves Lemma 5.1.

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College of Science, China University of Petroleum (Beijing), No. 18 Fuxue Road, Changping District, Beijing 102249, China

E-mail address: guoqilong1984@hotmail.com

School of Finance and Mathematics, Huainan Normal University, Huainan 232038, Anhui, P. R. China

E-mail address: cxyan@hnnu.edu.cn